

SPATIO-TEMPORAL ANALYSIS OF LAND USE LAND COVER AND EXPANSION OF URBAN AREA USING GIS AND REMOTE SENSING: A CASE STUDY ON CHINSURAH, HOOGHLY, WEST BENGAL

1. INTRODUCTION:

The demographic transformation of human societies into the urban era has pushed the monitoring of urban areas to the forefront of environmental and developmental agendas. Due to this transformation, a higher percentage of the world's population currently resides in urban areas than ever before, and growth in urban areas is occurring at an unprecedented rate. It has also increased socioeconomic disparities, crime, and vulnerability to natural and manmade risks. In this context, an assessment of spatiotemporal patterns of land-use, land-cover changes and the factors affecting these transformations is vital to developing rational, economic, social, and environment policies. Urban growth pattern has a direct influence on urban development processes both at the city and neighbourhood levels, but on the other hand growth pattern is also affected by land and housing policy, current built-up area, available developable area, change in household count, population, economy, and others. Remote sensing and GIS techniques can be used to analyse and model the urban growth pattern. Several statistical methods were applied to recognize and analyse this pattern, which shows increasing dispersed development of the city with the declining population growth rate. Apart from the derived results, this study also shows the potentiality of remote sensing and GIS techniques for the analysis of urban growth patterns.

2. STUDY AREA:

Chinsurah is a town of Hooghly district in West Bengal, located between 22° 56' 19" and 22° 52' 7" north latitude and between 88° 22' 1" and 88° 24' 14" east longitude. It has an area of 22.2927 square kilometers a total population of 177,833 (2011). It is the administrative headquarters of Hooghly.

Chinsurah was an important 17th century Dutch settlement. Although in 1656, the Dutch had built a factory there in Chinsurah and organized the town as their trading centre in 1795.

During the Napoleonic war, the settlement was occupied by a British garrison. Again during the peace process of 1814, Chinsurah was given back to the Dutch. But finally, Chinsurah, like other Dutch settlement in India were passed onto the British in

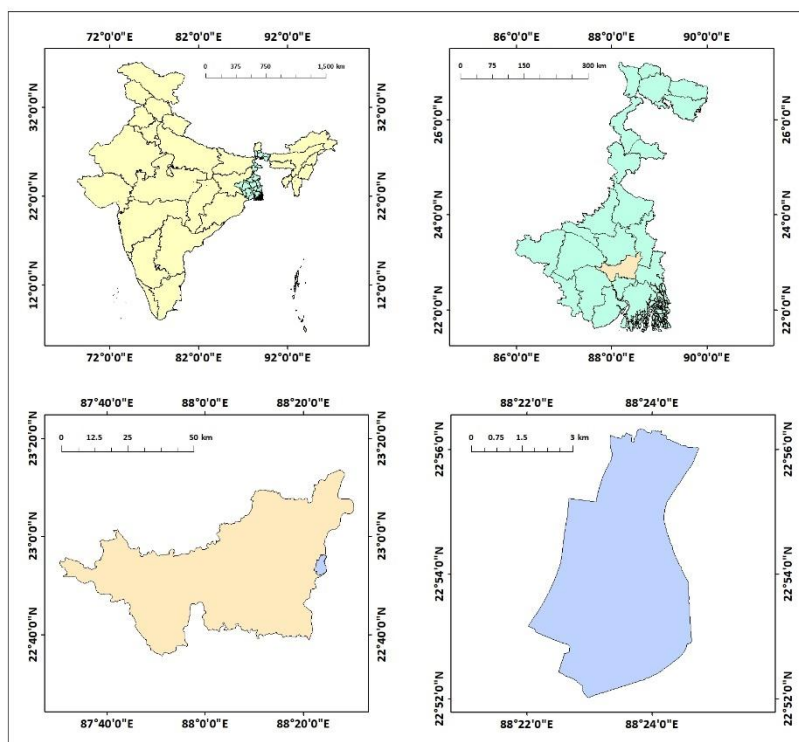


Figure 13: Location map of Chinsurah

1825 in exchange for their possession in Sumatra. The Portuguese, the Dutch, the French, the Danes and the British dominated industry, trade and commerce in this area for more than two centuries, and as a result this belt is highly industrialised. In Chinsurah municipality 64.87% of the population is urban and 35.13% is rural. The above map shows a portion of Chinsurah municipality.

3. DATA SOURCES AND METHODOLOGY:

Multi-temporal satellite data are used for this study. Landsat Thematic Mapper (TM-5) images of 2000 and 2007 and Operational Land Imager-8 (OLI-8) image for 2014 and 2021 are used for measurement of temporal urban growth pattern. The Landsat multi-temporal data are collected from the United States Geology Survey (<https://earthexplorer.usgs.gov/>).

The boundary map and district shapefile of Chinsurah town were obtained through geo referencing techniques. The collected images were pre-georeferenced to the UTM-Zone 45 North Projection with WGS-84 datum (Path/Row: 138/44).

SATELLITE	SENSOR	DATE OF AQUISION	SPATIAL RESOLUTION	RADIOMETRIC RESOLUTION	PATH	ROW
Landsat_5	TM	09/11/2000	30m	8 bit	138	44
Landsat_5	TM	12/11/2007	30m	8 bit	138	44
Landsat_8	OLI	16/11/2014	30m	16 bit	138	44
Landsat_8	OLI	19/11/2021	30m	16 bit	138	44

After making the LULC map of the study area the urban areas were extracted using ArcGIS software. For this extraction 'Raster Calculator' is used which can be found from the 'Map algebra' feature inside Spatial Analyst tool of the Arc toolbox. Then the extracted features are converted to vector layer using 'Raster to polygon' feature in Arc toolbox. Then the spatio-temporal pattern of urban expansion is calculated by using Symmetrical Difference tool. These maps are created in the temporal manner (2000, 2007, 2014, 2021).

After that contribution of LULC to the expanding urban area are calculated which is also known as urban encroachment. For this the 'clip' feature is used which can be found inside 'Geoprocessing' option in ArcGIS. At first the LULC map of beginning year is converted to polygon using 'Raster to Polygon' feature from Arc toolbox. Then the polygon is clipped with the extracted urban area of the ending year. Three maps are created from this method in Temporal manner (2000-2007, 2007-2014, 2014-2021).

Different metrics are calculated for showing ward wise distribution and pattern of urbanization between 2000-2021. The calculations have been done by using the following formulas-

$$\text{Urbanization Index (UI)} = \frac{\text{Total area of urban patch}}{\text{Total area of study area}} \times 100$$

$$\text{Urban Expansion Index (UEI)} = \left(\frac{\text{Urban area of ending year} - \text{Urban area of beginning year}}{\text{Time Span}} \right) \times 100$$

$$\text{Urban Expansion Intensity Index (UEII)} = \left(\frac{\text{Urban area of ending year} - \text{Urban area of beginning year}}{\text{Urban area of beginning year} \times \left(\frac{1}{\Delta t} \right)} \right) \times 100$$

$$\text{Annual Urban Expansion Rate (AUE)} = \left\{ \left(\frac{\text{Urban area of ending year}}{\text{Urban area of beginning year}} \right)^{\left(\frac{1}{\Delta t} \right)} - 1 \right\} \times 100$$

Urban Expansion Differentiation Index (UEDI) =

$$\left\{ \frac{\left(\frac{\Sigma \text{ward area of ending year} - \Sigma \text{ward area of beginning year}}{\Sigma \text{ward area of beginning year}} \right)}{\left(\frac{\text{Urban area of ending year} - \text{Urban area of beginning year}}{\text{Urban area of beginning year}} \right)} \right\} \times 100$$

Shapefile of Chinsurah (ward wise) is created and the above matrices are calculated through 'Field calculator' feature of the Attribute table. Then different thematic maps are made using 'Symbology' inside properties.

After that patch analysis for the urban areas are done in temporal manner. A fishnet is created and clipped with the extracted urban patches. Union operation is done between the urban polygon layer and the fishnet. For the patch analysis a plugin called 'Patch analyst' is needed in ArcGIS. With the help of 'Spatial statistics' option of patch analyst the map is generated which are then classified through the values of FD.

Now Urban hotspots are generated for the four years. 'Hotspot analysis with rendering' feature is used for the method which can be found in the rendering option in spatial statistics of Arc toolbox.

The Ellipse and mean centers are also generated to show the shifting directions of the urban areas between 2000-2021 using 'Directional distribution' and 'Mean center' feature from Arc toolbox.

4. RESULT AND DISCUSSION:

4.1 SPATIO TEMPORAL PATTERN OF URBAN EXPANSION:

The following maps are prepared to observe that how much urban area of Chinsurah is getting expanded in a temporal manner. The expansion is calculated by using Symmetrical Difference tool of ArcGIS Software.

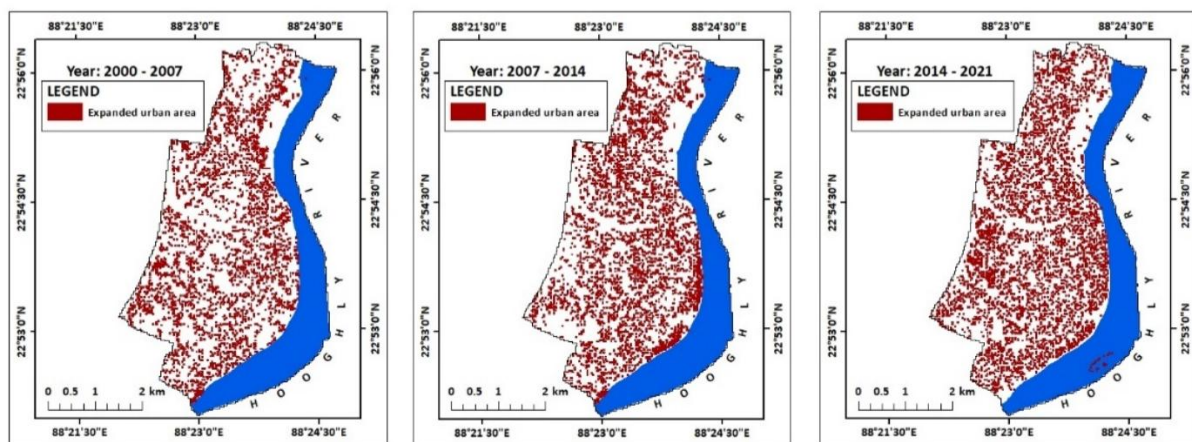


Figure 14: Expanded Urban Areas

From **2000-2007** the urban area expanded 3.3894 sq.km., from **2007-2014** the urban area expanded 3.8178 sq.km. and **2014-2021** the urban area expanded 4.1958 sq.km. So from here it can be seen that urban area is expanding in a rapid manner due to growth of different kinds of facility.

4.2 CONTRIBUTION OF LAND USE LAND COVER TO THE EXPANDING URBAN AREA IN TEMPORAL MANNER:

The following maps have been generated to get an idea about how much area of different land use land cover got encroached by the urban area. Or from another viewpoint it also can be said that the given bellow are describing which land use land cover are getting effected for the expansion of the urban area of Chinsurah municipality.

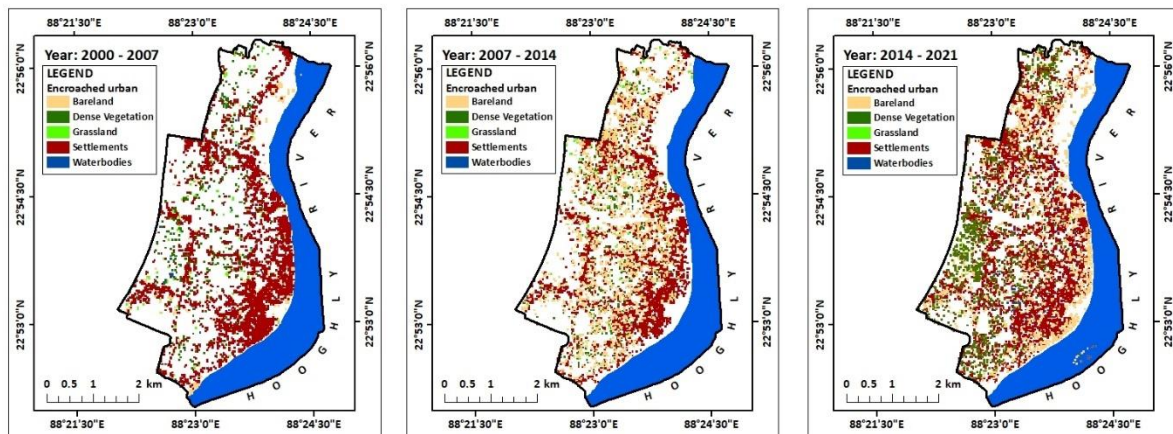


Figure 15: Encroached Urban Areas

In time span between **2000-2007**, the most affected area is Dense Vegetation (0.4599 sq.km.) and the area which got lesser affected is Waterbodies (0.0333 sq.km.). Apart from this Bare land area got encroached 0.3825 sq.km. and Grassland got encroached 0.1449 sq.km.

In time span between **2007-2014**, the most affected area is Bare land (1.8522 sq.km.) and the area which got lesser affected is Waterbodies (0.0009 sq.km.). Apart from this Dense Vegetation area got encroached 0.5238 sq.km. and Grassland got encroached 0.0927 sq.km.

In time span between **2014-2021**, the most affected area is Bare land (1.5147 sq.km.) and the area which got lesser affected is Grassland (0.0324 sq.km.). Apart from this Waterbodies area got encroached 0.1548 sq.km. and Dense vegetation got encroached 1.0557 sq.km.

So from here it can be seen that urban area is expanding in a rapid manner due to growth of different kinds of facility during 2000-2021

4.3 WARD WISE CALCULATION AND DISTRIBUTION OF MATRICES TO ENUMERATE THE PHENOMENON OF URBANIZATION:

By calculating matrices a temporal based change of a particular class can be detected. As here the temporal change of urban area of Chinsurah is being calculated. From which a vivid idea of the urban change can be understood.

4.3.1 Urbanization Index (UI):

In the year **2000** ward no. 20 and 30 has the highest urbanization Index, Whereas the wards located at centre and western parts have the lowest Urbanization index.

In the year **2007** ward no. 11 has the highest urbanization Index, Whereas the wards located at centre and entire western parts have the lowest Urbanization index.

In the year **2014** ward no. 28 has the highest urbanization Index, Whereas the wards located at northern and south western parts have the lowest Urbanization index.

In the year **2021** ward no. 11 has the highest urbanization Index, Whereas the wards located at northern and western and south western parts have the lowest Urbanization index.

4.3.2 Urban Expansion Index (UEI):

In between **2000-2007** ward no. 11 located at the eastern part has the highest Urban expansion Index, whereas the ward no. 22 and 30 located at south western parts have the lowest Urban expansion index.

In between **2007-2014** ward no. 8 located at the western part has the highest Urban expansion Index, whereas the ward no. 11 located at eastern part have the lowest Urban expansion index.

In between **20014-2021** ward no. 7 and 11 located at the eastern and central part has the highest Urban expansion Index, whereas the ward no. 8 located at western part have the lowest Urban expansion index.

Overall in between **2000-2021** ward no. 7 and 11 located at the eastern and central part has the highest Urban expansion Index, whereas the ward no. 22 and 30 located at south western part have the lowest

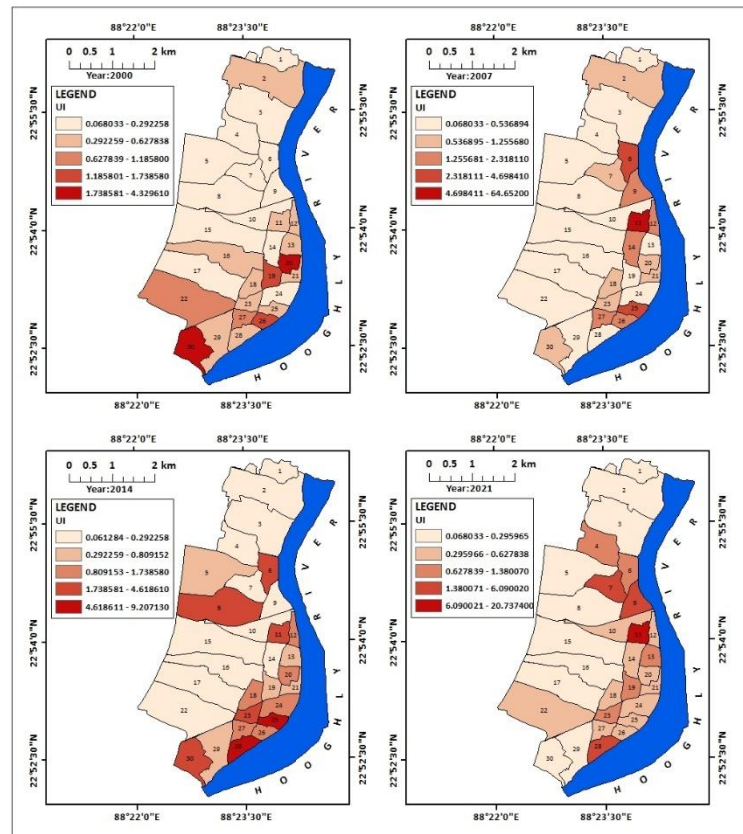


Figure 16: Ward wise maps for UI

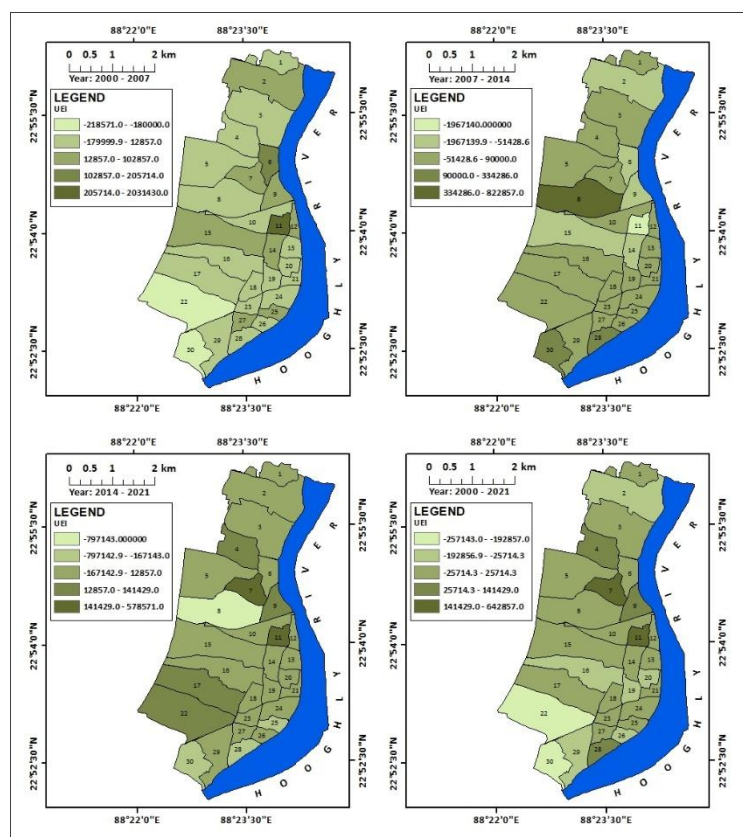


Figure 17: Ward wise maps for UEI

Urban expansion index.

4.3.3 Urban Expansion Intensity Index (UEII):

In between **2000-2007** ward no. 11 located at the eastern and central part has the highest Urban expansion intensity index, whereas the wards located at south and north western part have the lowest Urban expansion intensity index.

In between **2007-2014** ward no. 8 and 28 has the highest Urban expansion intensity index, whereas ward no. 2,6,7,9,11,14,15,21,27 have the lowest Urban expansion intensity index.

In between **2014-2021** ward no. 7 has the highest Urban expansion intensity index, whereas many wards located at northern and south eastern part have the lowest Urban expansion intensity index.

Overall in between **2000-2021** ward no. 11 located at the eastern and central part has the highest Urban expansion intensity index, whereas wards located at north, south, and central part have the lowest Urban expansion intensity index.

4.3.4 Annual Urban Expansion Rate (AUER):

In between **2000-2007** ward no. 11 has the highest Annual urban expansion index, whereas wards located at south western part have the lowest Annual urban expansion index.

In between **2007-2014** ward no. 8 and 28 has the highest Annual urban expansion index, whereas ward no. 2,9,11,14,15 have the lowest Annual urban expansion index.

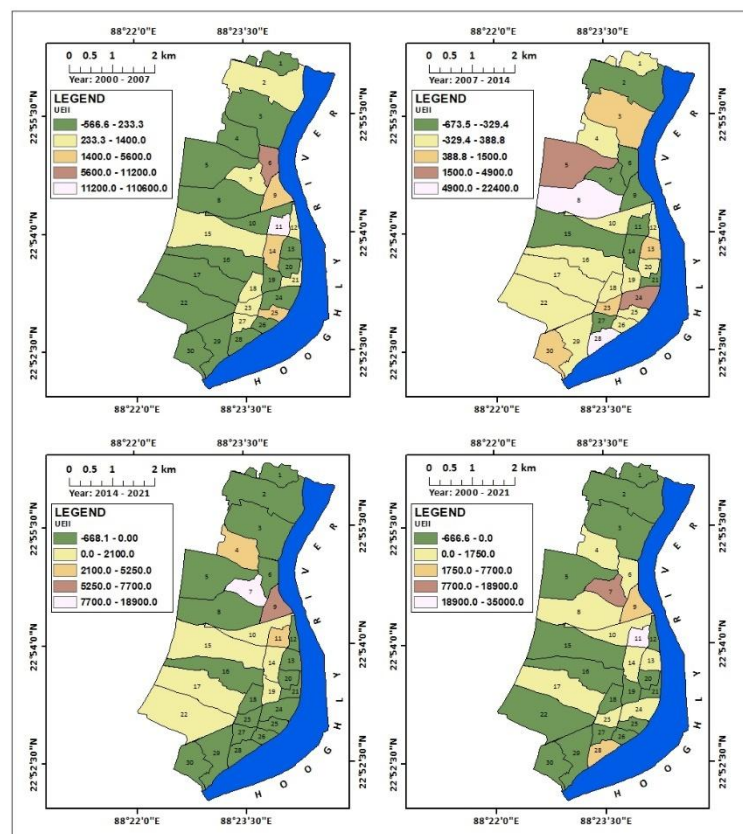


Figure 18: Ward wise maps for UEII

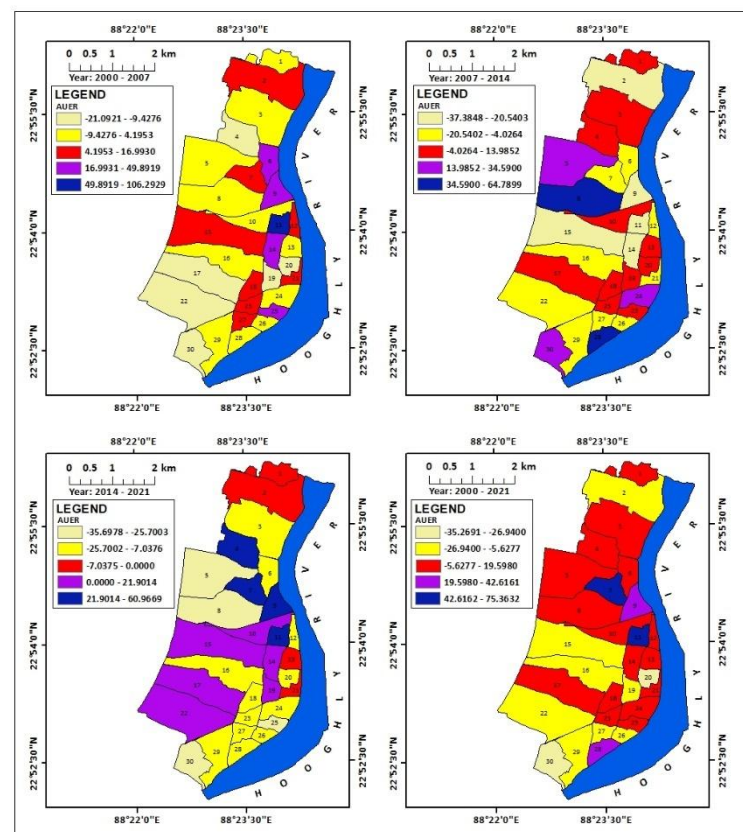


Figure 19: Ward wise maps for AUER

In between **2014-2021** ward no. 4,7,9 and 11 has the highest Annual urban expansion index, whereas ward no. 5,8,25 and 30 have the lowest Annual urban expansion index.

Overall in between **2021-2014** ward no. 7 and 11 has the highest Annual urban expansion index, whereas ward no. 20 and 30 have the lowest Annual urban expansion index.

4.3.5 Urban Expansion Differentiation Index (UED):

In between **2000-2007** ward no. 11 has the highest Urban Expansion Differentiation Index, whereas many wards in the north and southern parts have the lowest Urban Expansion Differentiation Index.

In between **2007-2014** ward no. 2, 7, 8, 9, 11, 14, 15, 21 and 27 has the highest Urban Expansion Differentiation Index, whereas ward no. 8 and 28 have the lowest Urban Expansion Differentiation Index.

In between **2014-2021** wards located at north, west and south eastern parts have the highest Urban Expansion Differentiation Index, whereas ward no. 7 has the lowest Urban Expansion Differentiation Index.

Overall in between **2000-2021** ward no. 11 has the highest Urban Expansion Differentiation Index, whereas many wards in the north, middle and southern parts have the lowest Urban Expansion Differentiation Index.

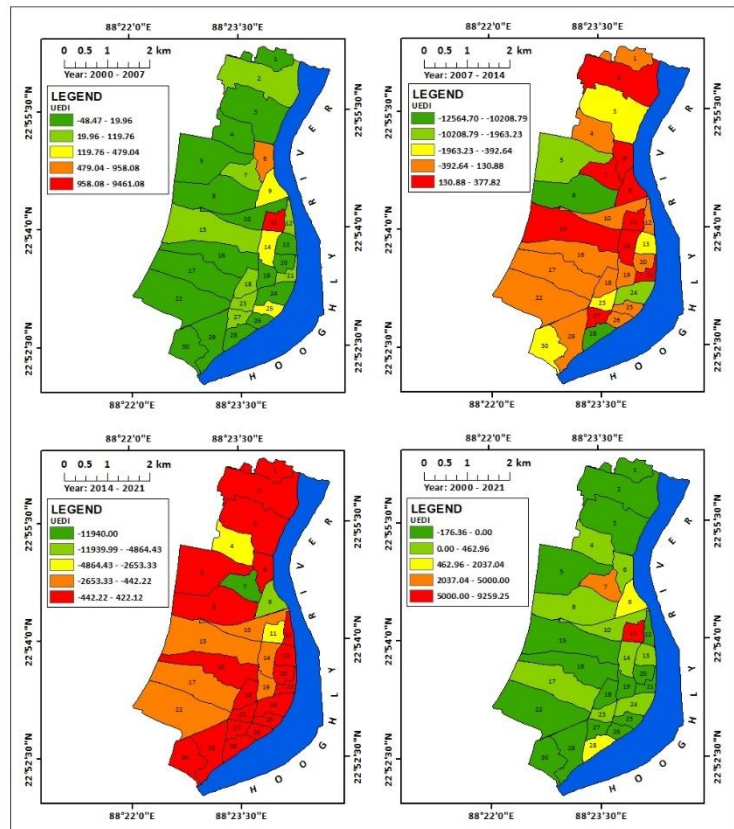


Figure 20: Ward wise maps for UEDI

These indexes can be used to evaluate the urban spatio-temporal expansion difference in a quantitative manner. Additionally, these could be employed to recognize the preferences of urban growth and to compare the speed or intensity of urban land use changes in a certain period.

4.4 SPATIO-TEMPORAL PATTERN OF FRAGMENTATION AND CLUSTERING OF URBAN PATCHES:

In the realm of urban planning and environmental management, Remote Sensing and GIS have emerged as indispensable tools for analysing complex urban landscapes. One powerful technique within this framework is patch analysis, which provides valuable insights into the spatial distribution and characteristics of land cover and land use patches within urban areas.

Patch analysis involves dividing the urban landscape into discrete units or patches based on criteria such as land cover type, size, shape, and spatial arrangement. By quantifying the spatial patterns of these patches, patch analysis enables planners and researchers to assess urbanization processes, monitor landscape changes over time, and inform sustainable development strategies. Here patch analysis is done using ArcGIS software on the basis of FD.

In the year **2000** the south eastern areas have highest level of clustering (1.28 – 1.36) and middle and western areas have highest level of fragmentation (1.49 – 1.60).

In the year **2007** the south eastern areas have highest level of clustering (1.30 – 1.36) and middle, north western and western areas have highest level of fragmentation (1.48 – 1.60).

In the year **2014** the north western areas have highest level of clustering (1.30 – 1.37) and middle and eastern areas have highest level of fragmentation (1.48 – 1.60).

In the year **2021** the north western areas have highest level of clustering (1.32 – 1.39) and middle and south eastern areas have highest level of fragmentation (1.50 – 1.60).

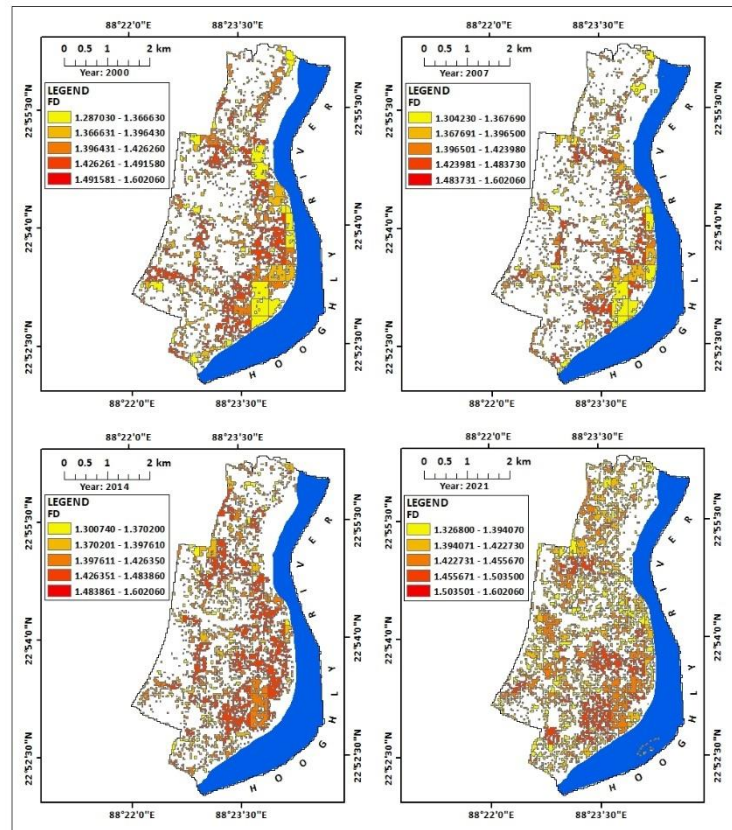


Figure 21: Maps for patch analysis of urban areas

4.5 DISTRIBUTION PATTERN OF URBAN PATCH FRAGMENTATION HOTSPOTS:

Hotspot analysis is a spatial analysis and mapping technique interested in the identification of clustering of spatial phenomena. These spatial phenomena are depicted as points in a map and refer to locations of events or objects. Generally two values are calculated first Bin value (which is a statistical kernel where different class interval or thresholds can be used) and the second one is Z Score (which is used for normalizing the data distribution).

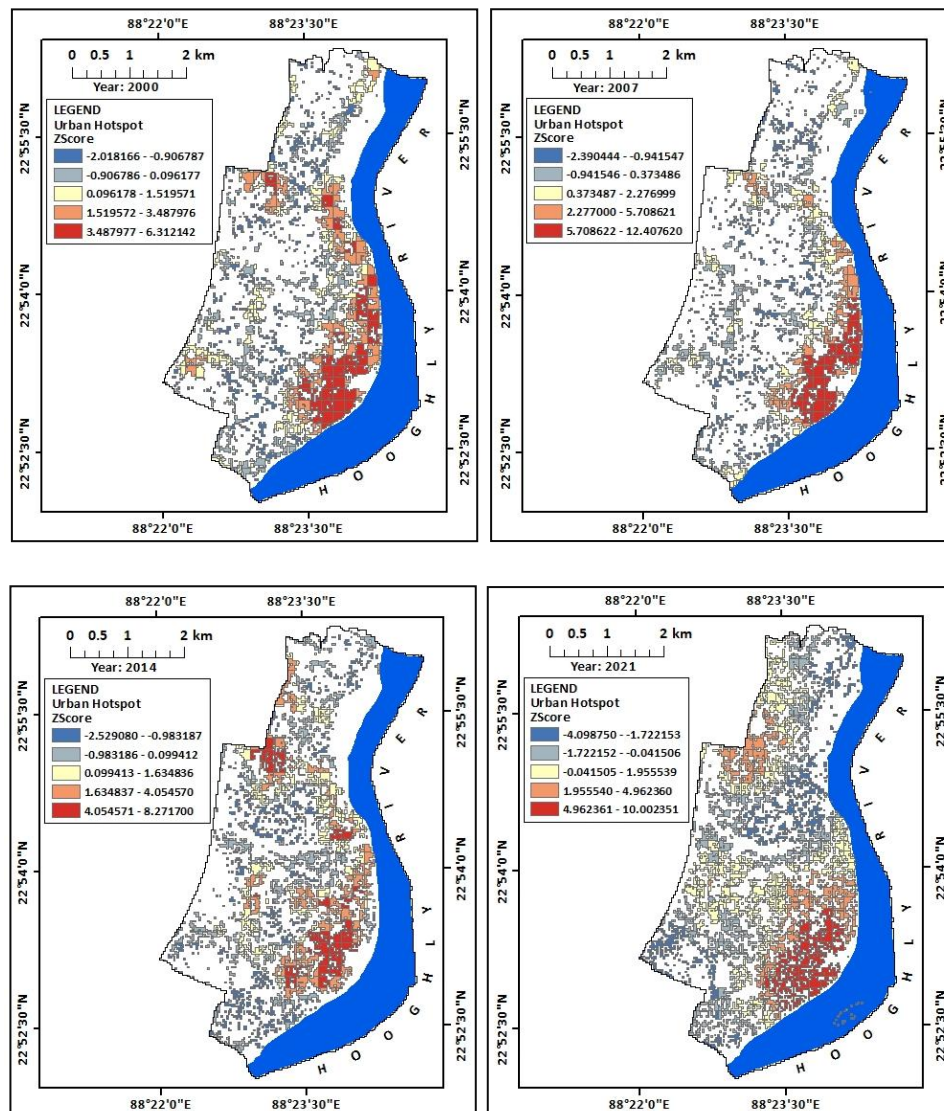


Figure 22: Hotspot analysis

For the calculation of the Hotspot of the urban patches as a weight field area have been chosen. As the maps displayed above are showing large hotspot areas indicates monotonical distribution (Grid wise) and homoscedastic variation of the urban patches. And this nature is being maintained though the urban area is expanding very rapidly. During **2000-2021** the highest values are seen in the south eastern regions, although in **2000** and **2014** some areas of north western region also shows highest value and lowest values are seen in the middle, west and northern regions.

4.6 Expansional Direction and Ellipse Plotting:

By calculating expansional direction, it can be estimated that in which direction mainly a particular class or portion is getting expanded or shifted on a temporal basis. Here the ellipses are being calculated based on 1 Standard Deviation where as a focus point weighted mean center is being used.

In **2000**, the urban is expanded mainly in 6.280013° .

In **2007**, the urban is expanded mainly in 5.923088° .

In **2014**, the urban is expanded mainly in 4.90086° .

In **2021**, the urban is expanded mainly in 6.649996° .

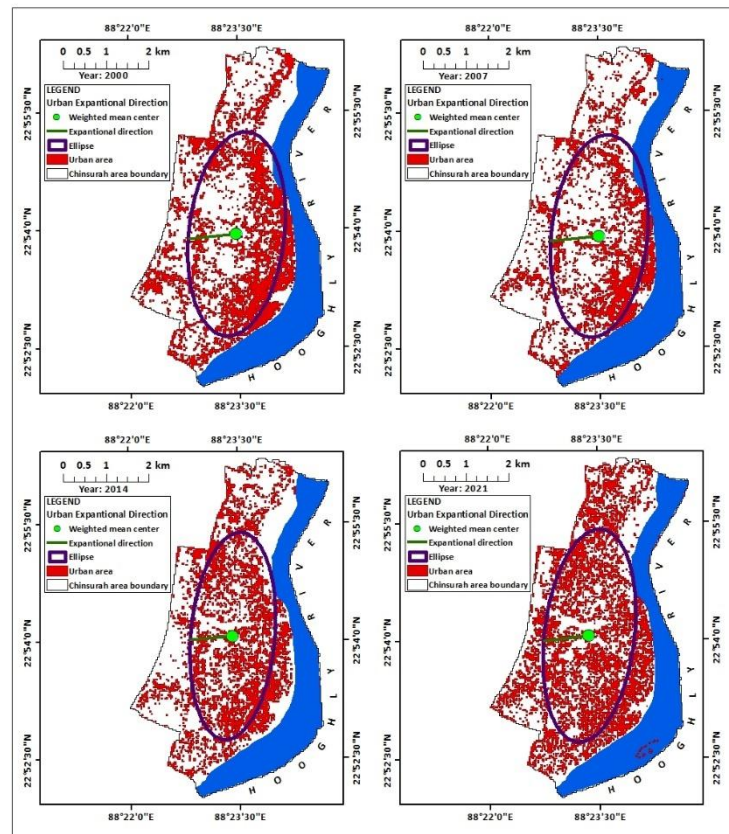


Figure 23: Expansional direction maps

5. CONCLUSION:

Urban organization indicates the diverse alignment of earthly objects and different land use characteristics which broadly include green space, built-up, water bodies and wetlands as well as open spaces in a region. So the perception and evaluation of urban dynamics in spatiotemporal context involve significant land transitions. From the above mentioned calculations we can see the urban regions expanding towards north western part of Chinsurah. Basically in this part, there is a large amount of bare land, which is getting filled day by day by urban growth.

Ultimately From this all analysis we can conclude that the spatial expansion rate of urban area in Chinsurah is high with expanding of facilities and that urban area will increase even more in future.

THEMATICAL REPRESENTATION OF LAND SURFACE TEMPERATURE (LST), SURFACE URBAN HEAT INDEX (SUHI) AND COMFORT ZONES: A CASE STUDY ON CHINSURAH, HOOGHLY, WEST BENGAL

1. INTRODUCTION:

The land surface temperature (LST) based surface urban heat island (SUHI) is suitable for exploring thermal environments at large scales. The surface urban heat island (SUHI) effect, which refers to the surface temperature difference between urban and rural sites, is one of the environmental impacts of urbanization. Compared to the air UHI effect, the land surface temperature (LST)-based SUHI effect is particularly suitable for monitoring thermal environments at a broad scale, like city/regional/continental/global scale. Land Surface Temperature (LST) is the result of how Earth's surfaces interact with the atmosphere, producing heat that affects people. It is influenced by meteorological aspects such as solar radiation, wind speed and surface properties. Surface urban heat island is a phenomenon where higher temperatures are experienced in the urban areas as compared to the surrounding rural or natural areas. Surface urban heat islands and high LST are primarily caused due to human activities and built environments, exhibiting an evident interconnectedness.

2. STUDY AREA:

This study selects Chinsurah municipality, the administrative headquarters of Hooghly district in West Bengal. It is located between 22° 56' 19'' and 22° 52' 7'' north latitude and 88° 22' 1'' and 88° 24' 14'' east longitude, with a total area of 22.2927 sq. km. and a total population of 177,833 (2011).

Based on weather reports collected between 1991 – 2021 average temperature of Chinsurah varies widely across seasons, varying between maximum 31°C and minimum 19°C, where May and January are the hottest and coldest months respectively. The temperature of May varies between 26 – 36°C and in January it is 13 – 15°C. The annual average rainfall is 1610.4mm. The wettest month is august with average 345.9mm precipitation.

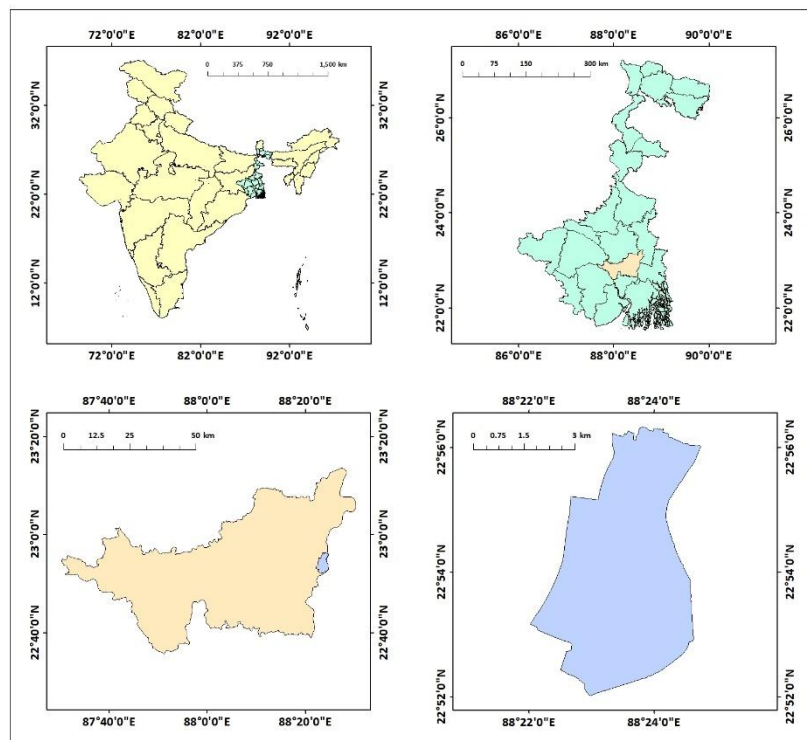


Figure 24: Location map of Chinsurah

3. DATA SOURCES AND METHODOLOGY:

For the present study we have considered data from the month of November which has the mean temperature 25°C and precipitation 28.9mm. Multiple satellite data are used for this study on temporal basis. Landsat Thematic Mapper (TM) images of 2000 and 2007 and Landsat-8 (OLI) images of 2014 and 2021 are used for measurement of LST, SUHI and Comfort zones. The Landsat multi-temporal data are collected from the United States Geology Survey (<https://earthexplorer.usgs.gov/>).

SATELLITE	SENSOR	DATE OF AQUISION	SPATIAL RESOLUTION	RADIOMETRIC RESOLUTION	PATH	ROW
Landsat_5	TM	09/11/2000	30m	8 bit	138	44
Landsat_5	TM	12/11/2007	30m	8 bit	138	44
Landsat_8	OLI	16/11/2014	30m	16 bit	138	44
Landsat_8	OLI	19/11/2021	30m	16 bit	138	44

After extracting the study area land surface temperature has been calculated using thermal bands of Landsat data. 'Raster Calculator' has been used to get the Radiance layer at first. Then LST is calculated using the Radiance layer. The formulas are given below -

$$\text{Radiance} = \text{RADIANCE_MULT_BAND} \times \text{Thermal Band} + \text{RADIANCE_ADD_BAND}$$

$$\text{LST} = \{ K2_CONSTANT_BAND / \ln(K1_CONSTANT_BAND + 1) - 273.15 \}$$

For SUHI the at first Z score is done with the LST layer and then using raster calculator the heat islands are extracted and converted into polygon. Mean centers are plotted on the basis of the heat islands to show the shifting trajectory.

Urban confort zones are determined by using Urban Thermal field Variance Index (UTFVI) with the help of Raster calculator. The formula is -

$$\text{UTFVI} = \frac{(\text{LST} - \text{Mean of LST})}{\text{Mean of LST}}$$

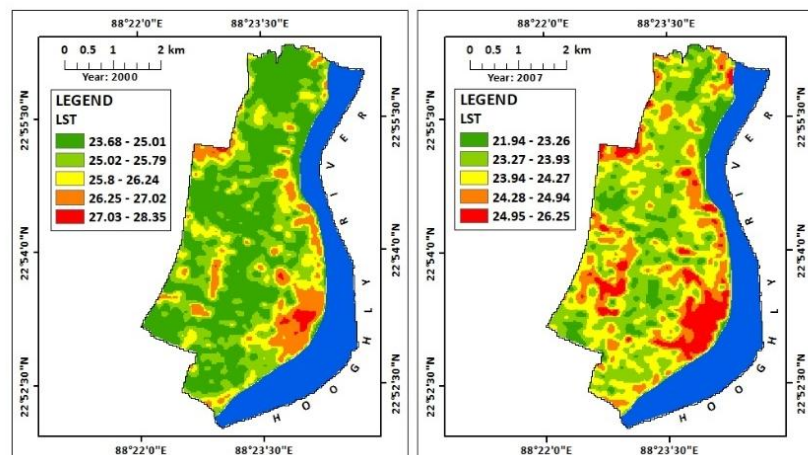
4. RESULT AND DISCUSSION:

4.1 LAND SURFACE TEMPERATURE (LST):

In **2000** the highest and lowest temperature is 28°C and 23°C respectively. Highest land surface temperature can be seen in a very little region of south eastern area of Chinsurah. The lowest temperature can be seen in a large area across the city, basically in the north and western portion.

In **2007** the highest and lowest temperature is 20°C and 26°C respectively. A larger area of highest temperature can be noticed at the south eastern, as well as west and north west areas of the study area. Areas with lowest temperature have become smaller and can be seen in many places across the city.

In **2014** the highest and lowest temperature is 21°C and 25°C respectively. The



highest temperature zones have become even larger and can be seen in the south east area and at some small areas in western portions. Lowest temperature can be noticed only in the north east areas.

In **2021** the highest and lowest temperature is 21°C and 26°C respectively. Highest temperature areas are largest compared to previous 3 years. It can be seen in south east, west and north west regions. There is a very small area with lowest temperature at north east area.

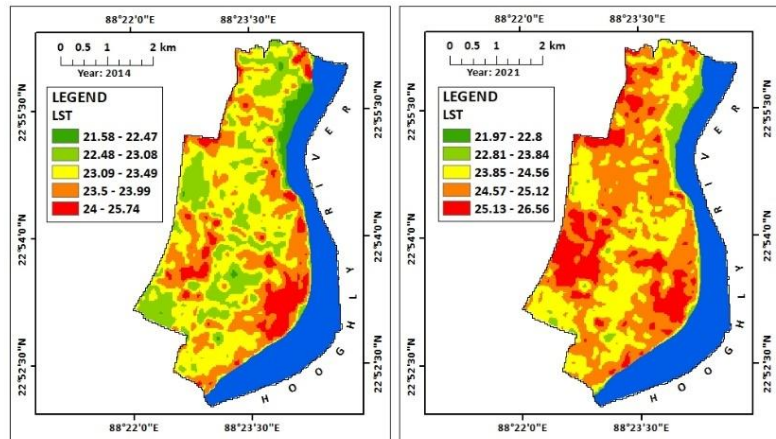


Figure 25: Maps for LST

4.2 SURFACE URBAN HEAT ISLAND (SUHI):

In **2000** there is a very small area of SUHI at the south eastern part of study area. The mean center is located at the center of the study area. The total SUHI area is 0.1809 sq.km.

In **2007** the area of SUHI has increased and the mean center has shifted towards north. The total SUHI area is 1.4373 sq.km.

In **2014** the area has become larger and it has shifted towards west a little bit. The total SUHI area is 1.6758 sq.km.

In **2021** the SUHI area is largest compared to previous years and again the area has shifted towards west. The total SUHI area is 3.6954 sq.km.

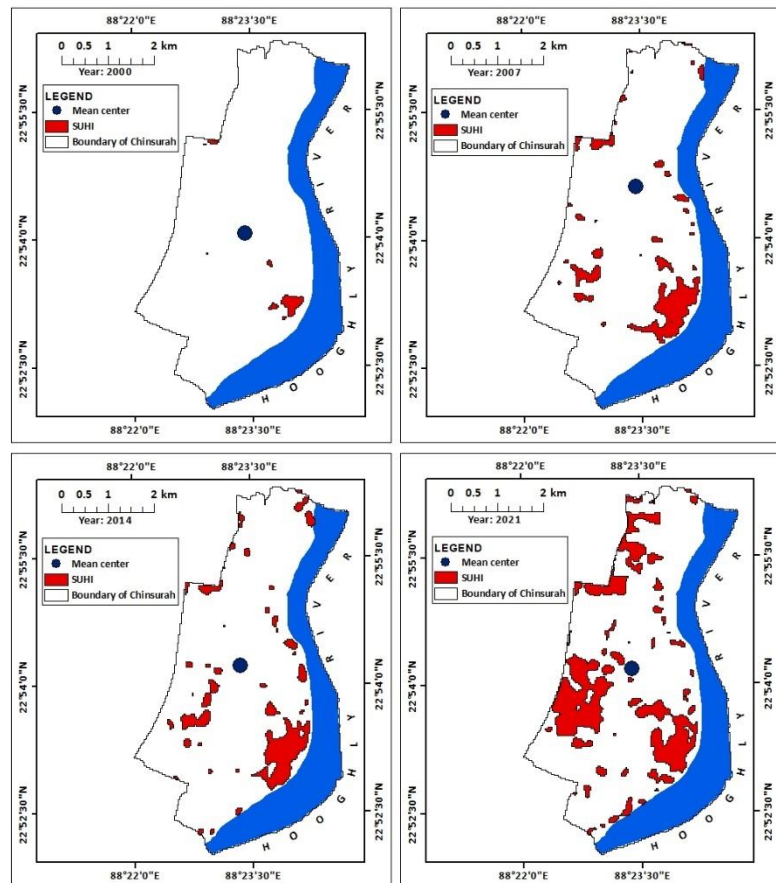
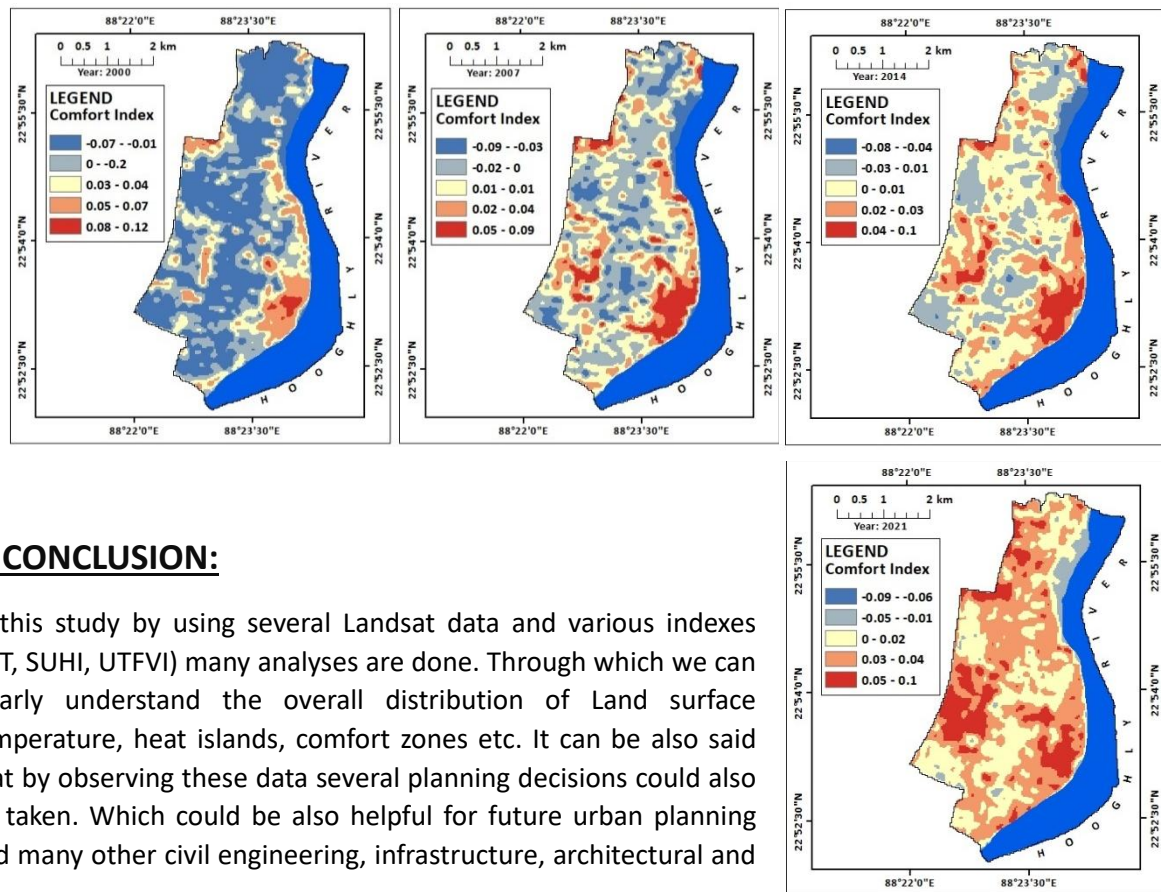


Figure 26: Maps for SUHI

4.3 COMFORT ZONES:

In all of the maps it is clear that the most comfortable zones are where the LST is lowest and least comfortable zones are the areas which is located on the heat islands. The values of comfort index

varies between -0.1 to 0.1 , where the lesser value indicated more comfortness and higher value indicates less comfortness.



5. CONCLUSION:

In this study by using several Landsat data and various indexes (LST, SUHI, UTFVI) many analyses are done. Through which we can clearly understand the overall distribution of Land surface temperature, heat islands, comfort zones etc. It can be also said that by observing these data several planning decisions could also be taken. Which could be also helpful for future urban planning and many other civil engineering, infrastructure, architectural and even defence and intelligence-related applications and uses for Chinsurah.

Figure 27: Maps for Comfort Zones

SPATIAL DISTRIBUTION OF THE SEASONAL GROUND WATER DEPTH ON TEMPORAL BASIS: A CASE STUDY ON PASCHIM MEDINIPUR, WEST BENGAL

1. INTRODUCTION:

Human civilization depends heavily upon the water. Almost 97% of the total available water on the earth is saline by nature and rest of the 3% remains as freshwater. Most of the freshwater is stored as ice in glaciers and polar ice sheets. Almost 30% of total freshwater is stored in aquifers as groundwater and less than 1% is available at lakes and rivers. With an extensive growth in population, there is a tremendous pressure on freshwater resources all over the world, especially in the developing countries. In India, demand for water is growing at an alarming rate. India is heavily dependent on groundwater for various purposes. Groundwater provides 80-90% of domestic water supply in rural areas. A total of 50% of urban and industrial demand is also dependent on groundwater. Groundwater level is declining due to excessive ground water extraction. The exploitation of groundwater has taken place at a speed which does not allow the water table to recover its losses. Recharge process is inadequate in comparison of the rate of extraction. As a result of this, shallow aquifers are drying up and drought-like situation is happening over large parts of the country during pre-monsoon season. Thus, identification of groundwater potential Zones is very important for the optimum utilization and conservation of this precious resource.

2. STUDY AREA:

Paschim Medinipur in West Bengal, India lies between 21°46' N to 22°57' N latitudes and 86°33' E to 87°44' E longitude, covered with an area of 9081.13 sq. km. The district enjoys with tropical climate and the land surface of the district is characterized by hard rock uplands, lateritic covered area, and flat alluvial and deltaic plains. Extremely rugged topography is seen in the western part of the district and rolling topography is experienced consisting of lateritic covered area. These rolling plains gradually merge into flat alluvial and deltaic plains to the east and south east regions of the district. The soil is fairly fertile and a great group of the Taxonomic classification system as paleustalfs and haplaquents. Average temperature of the district varies widely across seasons, varying between maximum 43°C and minimum 7°C respectively with an annual average rainfall, 1428 mm.

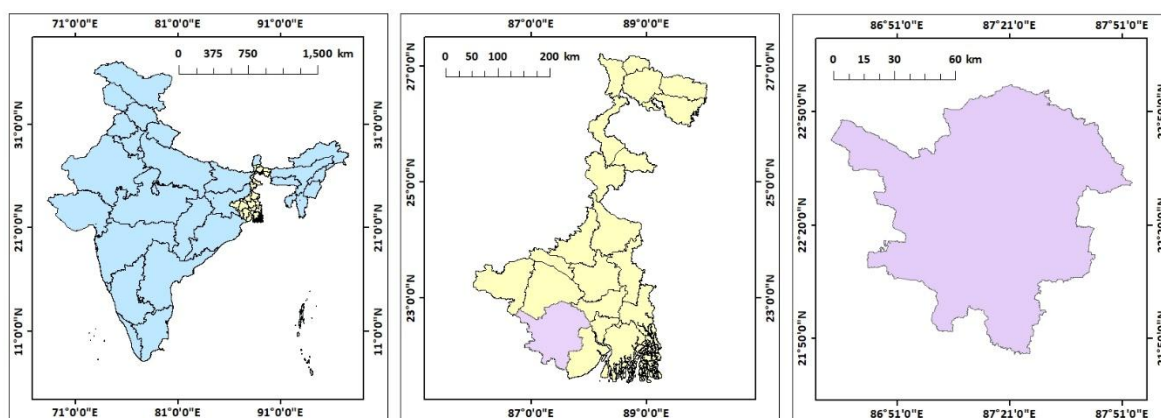


Figure 28: Location map of Paschim Medinipur

3. DATA SOURCES AND METHODOLOGY:

The data that have been used for the present work are as follows –

DATA	SOURCES
DEPTH POINTS	https://indiawris.gov.in/wris/#/

On the basis of the downloaded data fluctuations are calculated using MS Excel. On the basis of the data depth points are plotted and thematic maps are made by using IDW interpolation feature of ArcGIS.

4. RESULT AND DISCUSSION:

Groundwater level of pre-monsoon, monsoon and post-monsoon reflects the actual groundwater depth distribution and fluctuation of the area on a temporal basis. Groundwater depth data of different observatory wells for the time period of **1996,2003,2010,2017** has been collected from the India Water Resources Information System (WRIS).

4.1 SPATIAL DISTRIBUTION OF GROUND WATER DEPTH (SEASONALLY AND TEMPORALLY):

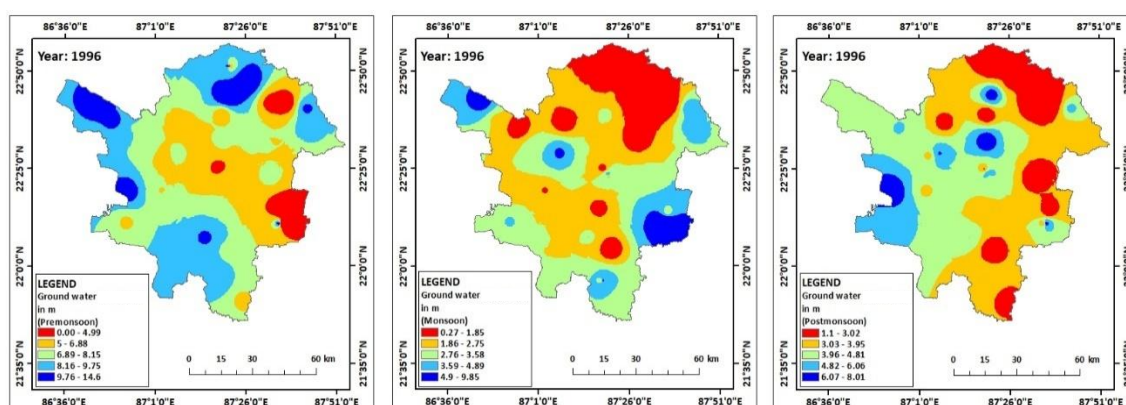


Figure 29: Seasonal ground water depth maps of 1996

The above maps are describing the distribution of Ground water seasonally (pre monsoon, monsoon, post monsoon) in **1996**. Which means in pre monsoon ground water level varies from 0 – 14m, in monsoon it is 0 – 10m and in post monsoon it is 1 – 8m below ground.

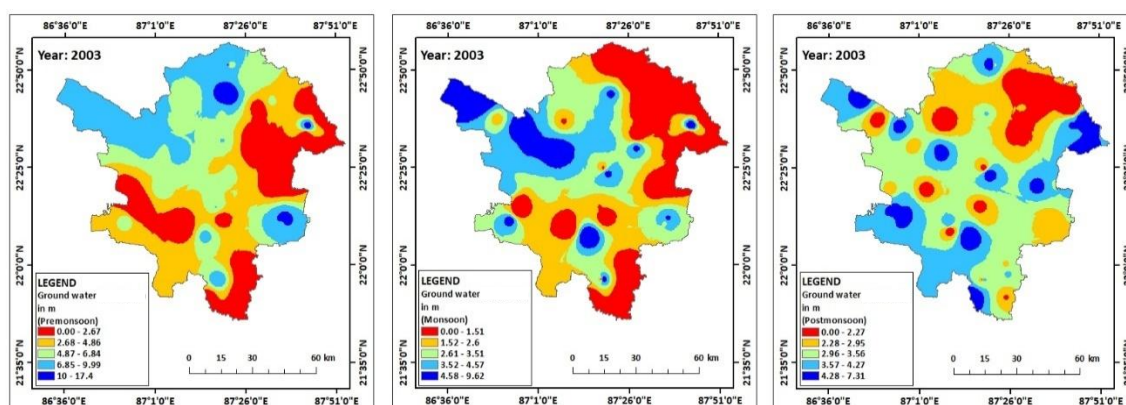


Figure 10: Seasonal ground water depth maps of 2003

The above maps are describing the distribution of Ground water seasonally (pre monsoon, monsoon, post monsoon) in **2003**. Which means in pre monsoon ground water level varies from 0 – 17m, in monsoon it is 0 – 10m and in post monsoon it is 0 – 7m below ground.

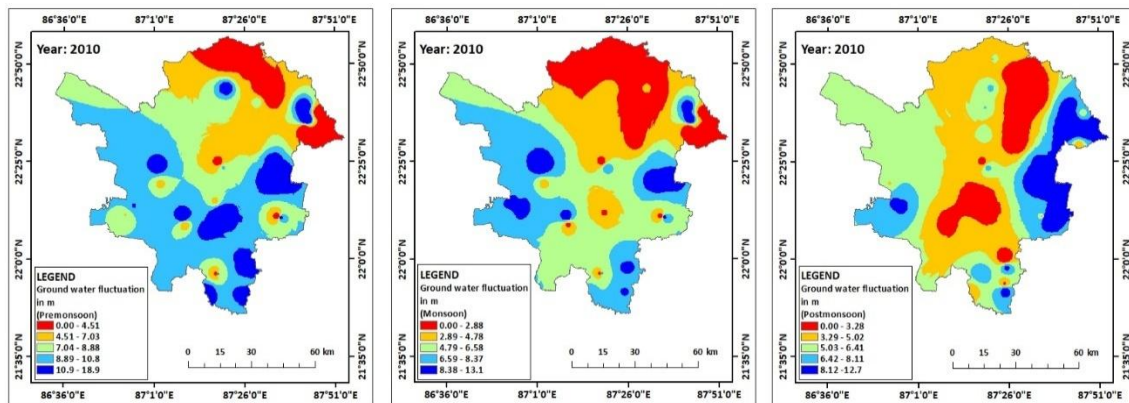


Figure 31: Seasonal ground water depth maps of 2010

The above maps are describing the distribution of Ground water seasonally (pre monsoon, monsoon, post monsoon) in **2010**. Which means in pre monsoon ground water level varies from 0 – 19m, in monsoon it is 0 – 13m and in post monsoon it is 0 – 12m below ground.

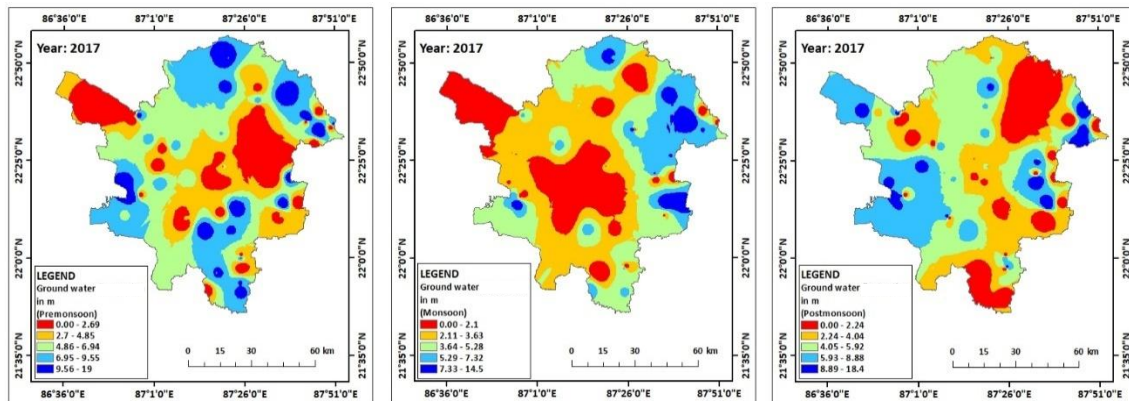


Figure 32: Seasonal ground water depth maps of 2017

The above maps are describing the distribution of Ground water seasonally (pre monsoon, monsoon, post monsoon) in **2017**. Which means in pre monsoon ground water level varies from 0 – 19m, in monsoon it is 0 – 14m and in post monsoon it is 0 – 18m below ground.

4.2 DISTRIBUTION OF AREAS HAVING SEASONAL FLUCTUATION OF GROUND WATER:

The ground water fluctuation map describes that, in **1996** the highest fluctuation can be seen in the north region (Garhbeta). The lowest fluctuation can be seen at the middle part and very few parts of north east, east, south east and western parts of Medinipur (Some portions of Kharagpur, Jhargram etc.).

In **2003** the highest fluctuation can be seen in the very small area of north region (Salboni). The lowest fluctuation can be seen across a large area of entire south, middle, south west and eastern region.

In **2010** the highest fluctuation can be seen in the north region (Garhbeta). The lowest fluctuation can be seen at mainly at east regions and small areas of west, south and south regions.

In **2017** the highest fluctuation can be seen in the small areas of south eastern regions. The lowest fluctuation can be seen at a large part of middle area as well as small areas of north west, south east, and south regions.

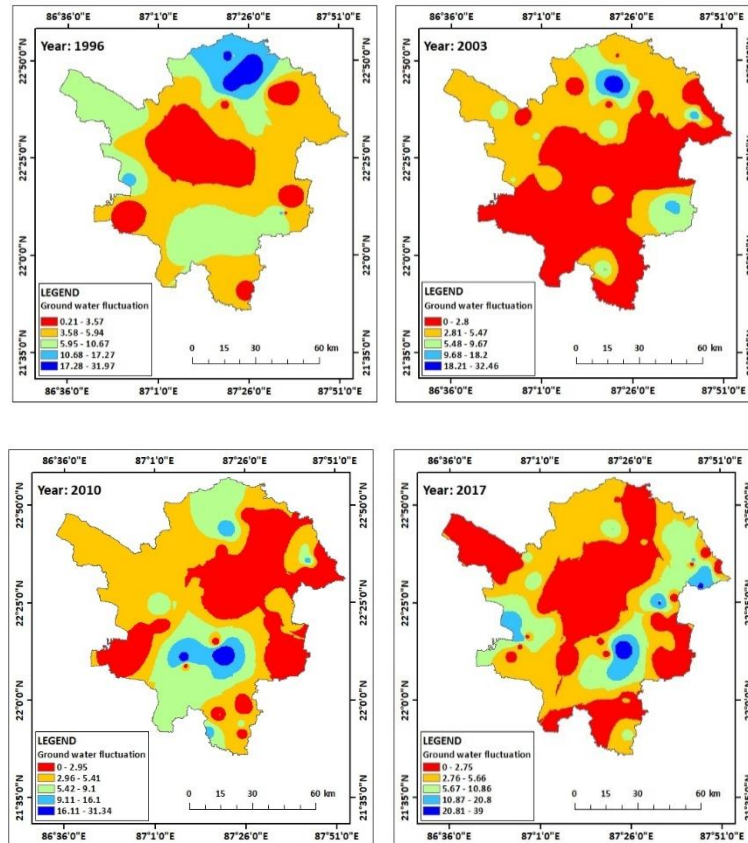


Figure 33: Distribution of Areas Having Seasonal Fluctuation of Ground Water

5. CONCLUSION:

The present study attempts to identify the groundwater potential zones in one of the water-stressed districts of West Bengal. Delineation has been done through weighted overlay model by integrating RS GIS. The study reveals that 16.53% of the total area is under the low potential zone, it is mainly due to impervious geological setting and high slope. Major portion of the study area (60.92%) falls under moderate groundwater potentiality zones and rest of the area (22.55%) is under the high potential zone. It is clear that pourous geological setting, low slope, permeable soil texture coupled with higher lineament density and vegetative cover helps in the development of the high potential zone. The applied approach has merits and can be used elsewhere effectively with suitable modifications. This study can help the Concerned decision makers to formulate an effective plan for the study area.

TEMPORAL ANALYSIS OF FOREST COVER MAPS AND DEGENERATION OF FOREST USING GIS AND REMOTE SENSING: A CASE STUDY ON JHARGRAM, WEST BENGAL.

1. INTRODUCTION:

Deforestation is one of the most pressing environmental issues that the world is facing currently. It is the conversion of forested land to non-forested land by humans. Deforestation occurs when a land dominated by naturally occurring trees is converted to provide certain services in response to the human demand. The indiscriminate felling of trees has resulted in a reduction of 3.16% in the global forest cover from 1990 to 2015. Although India has seen an increment in the total forest cover of ca. 1%, still there are certain regions in the country that have sought a decrease in the forest cover. The main reasons attributed to the reduction in forest cover are shifting cultivation, diversion of forest lands for developmental activities, etc. Continuous cutting of trees has impacted the microclimatic conditions, hydrological cycle, soil quality, biodiversity, etc. of the country, thereby making the country more vulnerable for any uneventful happening. Sustainable forest management practices, alternatives for shifting cultivation, promotion of plantation outside the forest and the usage of certified forest products, etc. are some of the measures that can be adopted to curb the rate of deforestation.

2. STUDY AREA:

Jhargram is a district in the state of West Bengal, India. The district lies between the Kangsabati River in the north and the Subarnarekha in the south. Jhargram has one of the lowest population densities among the districts of West Bengal, with almost all its population living in rural areas. It is a popular tourist destination known for its sal forests, elephants, ancient temples and royal palaces. The district was formed on 4 April 2017, after bifurcation from the Paschim Medinipur district as the 22nd district of West Bengal. The district has its headquarters at Jhargram.

Jhargram has an average elevation of 81 metres (265 feet). The weather, like much of Bengal, is extremely humid and tropical. Temperatures can reach as high as 46 °C in the hot and dry months of May and June but it can be up to 4 °C in the chilly nights of December and January.

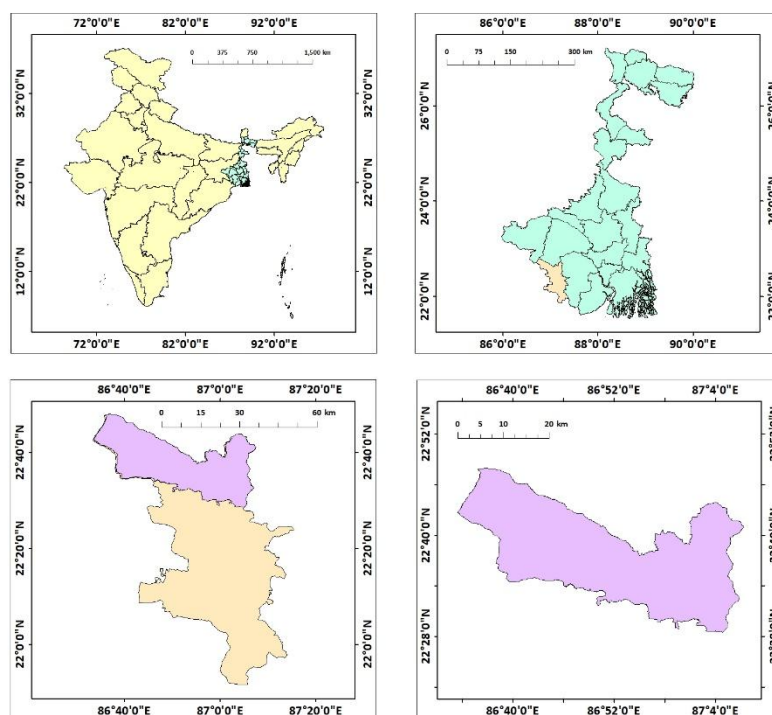


Figure 34: Location map of Binpur

Presently the forests of Jhargram Division are situated in the Civil Sub-division of Jhargram of Jhargram District and cover the Civil Blocks viz. Binpur-I (the portion on the West of the Kangsabati river), Binpur-II, Jhargram, Jamboni, Gopiballavpur-I & Gopiballavpur-II and police stations of Belpahari, Binpur, Jamboni, Jhargram, Gopiballavpur and Beliyabera. The Jhargram Forest Division lies between 21°52' and 22°48' North latitudes and 86°34' and 87°20' East longitude approximately. On the North, it is bordered by the civil districts of Purulia and Bankura and on the East, it is bordered by the river Kangsabati (from the western border of Midnapore Division) and partly by the river Subarnarekha from the western border of Kharagpur Division. Jharkhand shares a border with the State of Orissa in the South and in the West with the Jharkhand State. In the present study Binpur sub-division of Jhargram district has been selected for the following analysis.

3. DATA SOURCES AND METHODOLOGY:

For the present study, to understand the pattern of deforestation and afforestation in the Jhargram district of West Bengal Landsat Thematic Mapper (TM-5) images of 1993, 2000 and 2007 are used. The Landsat multi-temporal data are collected from the United States Geology Survey (<https://earthexplorer.usgs.gov/>).

SATELLITE	SENSOR	DATE OF AQUISION	SPATIAL RESOLUTION	RADIOMETRIC RESOLUTION	PATH	ROW
Landsat_5	TM	04/04/1993	30m	8 bit	139	44
Landsat_5	TM	21/03/2000	30m	8 bit	139	44
Landsat_5	TM	25/03/2007	30m	8 bit	139	44

At first the Radiance and TOA has been calculated for band no. 1(Blue), 3(Red) and 4(NIR). With the help of those TOA bands EVI (Enhanced Vegetation Index) has been calculated to make the forest cover maps. All these works have been done using the raster calculator. The formulas for Radiance, TOA and EVI are -

- Radiance (L_λ) = $\left(\frac{LMAX_\lambda - LMIN_\lambda}{Q_{cal\ max} - Q_{cal\ min}} \right) \times (Q_{cal} - Q_{cal\ min}) + LMIN_\lambda$
- TOA (Top of Atmosphere) (P_λ) = $\left(\frac{\pi \times L_\lambda \times d^2}{ESUN_\lambda \times \cos \theta_s} \right)$
- EVI (Enhanced Vegetation Index) = $2.5 \times \left\{ \frac{(NIR - RED)}{(NIR + 6) \times (RED - 7.5) \times (BLUE + 1)} \right\}$

After making the forest cover maps 2 more maps have been created to show the forest degeneration (1993 – 2000, 2000 - 2007). To get the forest degeneration map union operation has been done between 2 forest cover maps.

Hotspot areas of forest degeneration has been generated with the help of “Hotspot analysis with rendering” option of arc toolbox.

4.RESULT AND DISCUSSION:

4.1 PREPERATION OF FOREST COVER MAP OF THE STUDY AREA:

The forest cover map of the Binpur sub-division has been made according to the above methodology. Forest cover maps are made in temporal basis (1993, 2000 and 2007). In 1993 the forest area was 175 sq km. approximately which in 2000 decreased to 83 sq km and in 2007 it became 69 sq km approximately. The forest cover maps are given below-

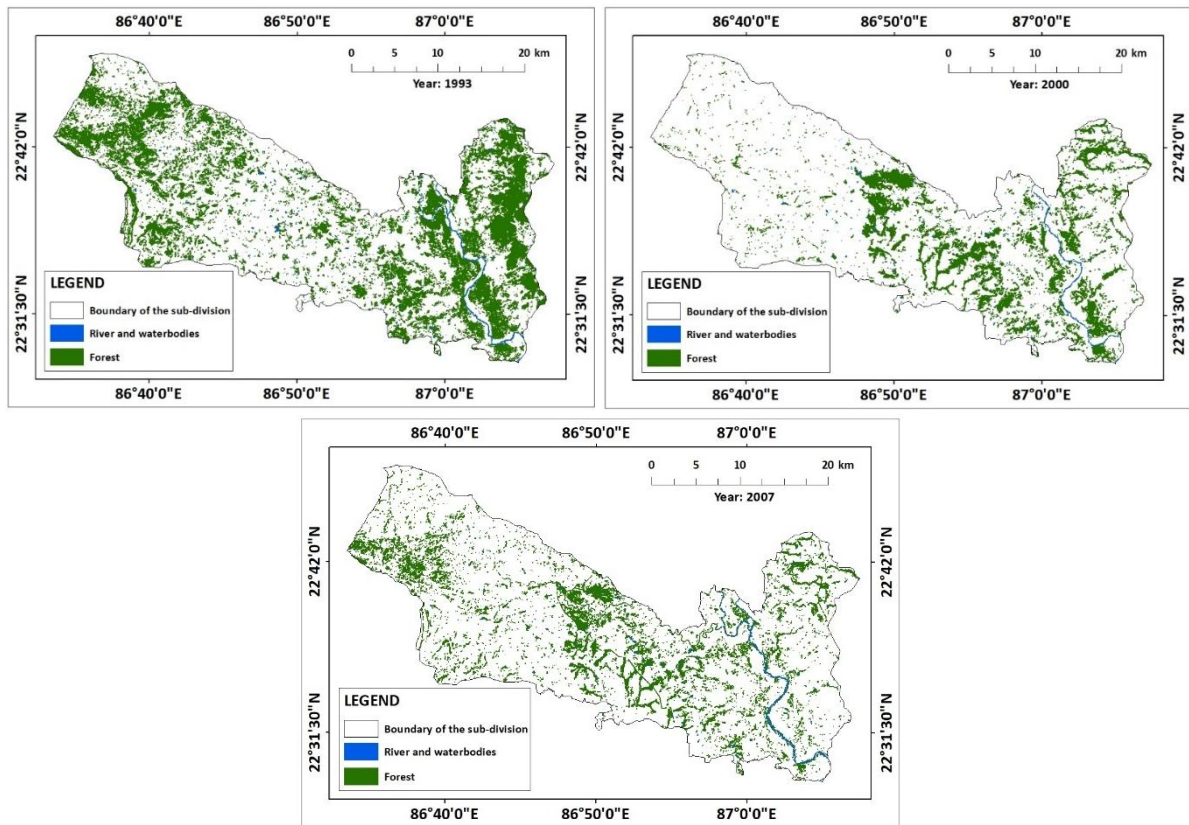


Figure 35: Forest cover maps of 1993, 2000 and 2007

4.2 DEMARCATION OF THE AREAS HAVING FOREST DEGENERATION:

The following maps have been made on the basis of forest degeneration. From **1993** to **2000** the amount of forest degeneration was much higher compared to the amount of forest degeneration between **2000** to **2007**. The amount of forest loss between year 1993 – 2000 is much higher than the time between 2000 – 2007.

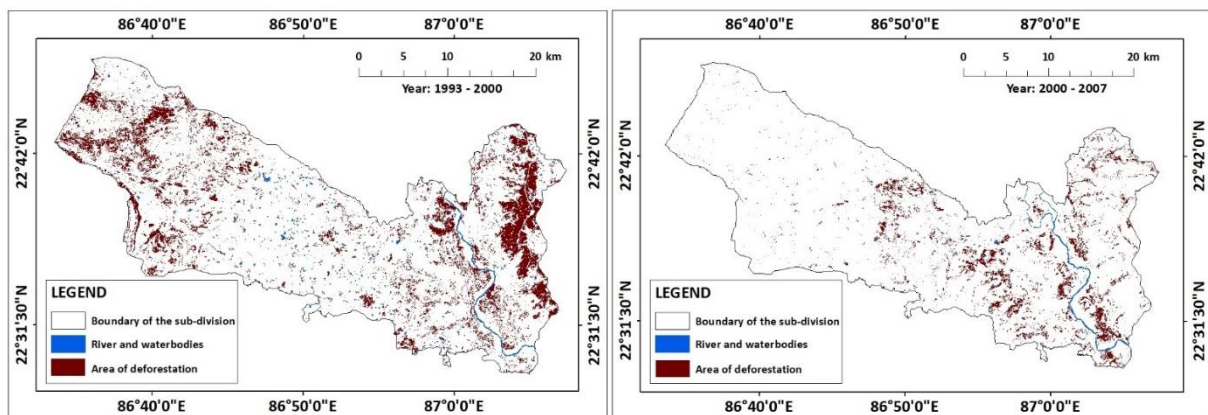


Figure 36: Forest Degeneration Area

4.3 AREAS OF BINPUR SUB-DIVISION HAVING DEGENERATION HOTSPOT:

Areas having forest degeneration hotspots have been calculated on the basis of z-score. Between **1993 - 2000** highest hotspots can be seen in the eastern and western regions of Binpur and whereas between **2000 – 2007** the hotspots can be noticed in the middle regions and around the river located in the east.

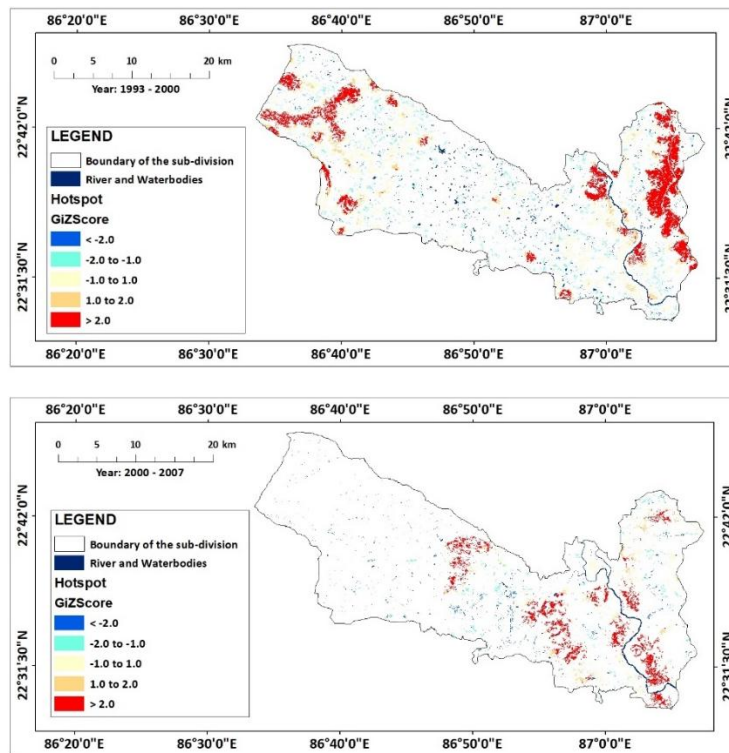


Figure37: Degeneration Hotspots

5. CONCLUSION:

Deforestation is a global issue and is prevalent throughout the geographic range of many primate species. Although deforestation has occurred for tens of thousands of years, deforestation in the twentieth and twenty-first centuries was primarily driven by agricultural crop expansion, cattle farming, fire, logging, mining for gas, oil, or minerals, transportation, and urban development.

The temporal analysis of the study area reveals that the forest area of north Jhargram (Binpur) is always changing in nature and pattern and there is a contribution of the expanding urban areas in the degeneration of the forests.